LECTURE L13-14 · MODULE 2 · CO2

Visual storytelling

Week 7 | Slides: `DATA209_Module_2_Core_Advanced_EDA.pptx`, pages 30–36 | Laboratory: **P13-14 (Assignment 1)**

TOPICS COVERED

- Grammar of graphics
- Good vs. misleading plots
- Visualising non-numeric data
- Dashboard logic and storytelling with data

The grammar of graphics

A plot is built from layers. Thinking in layers is what lets you answer "which plot?" by reasoning rather than by recall.

1. **Data** — the tidy table being shown, one row per observation.
2. **Aesthetics** — the mapping from variables to visual properties: x, y, colour, size, shape.
3. **Geometries** — the marks that carry the data: points, bars, lines, boxes, tiles.
4. **Scales** — how data values become visual values: axis limits, log scales, colour palettes.
5. **Facets** — small multiples; the same plot repeated across the levels of a variable.
6. **Coordinates and theme** — coordinate system plus the non-data ink: labels, grid, legend.

Choosing a chart

YOU HAVE	QUESTION	PLOT
one numeric	what is its shape?	histogram, density, boxplot, ECDF
one categorical	how common is each level?	bar plot of counts or proportions
two numeric	do they move together?	scatterplot, plus a smoother
numeric + categorical	do the groups differ?	grouped boxplot, violin, overlaid densities
two categorical	are they associated?	cross-tab, stacked bar of proportions, mosaic
many numeric	where is the structure?	correlation heatmap, pair plot, parallel coordinates
numeric over time	what is the trend?	line chart, ordered by time

Pie charts appear nowhere in this table. They ask the reader to compare angles, which people do poorly; a bar chart beats them in almost every case.

Six ways a plot misleads

- **Truncated axis** — a bar chart not starting at zero exaggerates small differences.

- **Dual axes** — two scales on one chart manufacture correlation by choice of range.
- **3-D and area encodings** — doubling a radius quadruples the perceived size.
- **Cherry-picked range** — where the time axis starts decides whether the trend rises or falls.
- **Rainbow colour maps** — non-perceptual palettes create bands that are artefacts of the colours.
- **Hidden aggregation** — plotting means without spread or sample size conceals both.

The test to apply to your own work: *could someone reach the opposite conclusion from this data with a different but equally reasonable choice?*

Non-numeric data

Ordered bar plots for ordinal data — never let the software sort alphabetically. Stacked bars of proportions for part-to-whole comparisons. Mosaic plots when both variables are categorical with several levels. Heatmaps for contingency tables, using a perceptually uniform colour scale. For high cardinality, plot the top 15 and aggregate the rest into a labelled "Other".

Assignment 1 preparation

The mini-project due this week is marked on four criteria: **technique 30%, interpretation 35%, reproducibility 15%, presentation 20%**. Interpretation carries the most weight, which is deliberate — it is the skill the course exists to build.

What to bring to the presentation:

1. The problem statement and what a row represents, with the sampling frame stated.
2. The dependent/independent split, justified.
3. Univariate findings: shape, centre, spread and anomalies per variable.
4. Bivariate findings, including at least one relationship you did not expect.
5. Multivariate structure — grouping or clustering, interpreted in plain language.
6. Three hypotheses, each with a falsification condition and its confounders named.

Two failures cost the most marks: **figures with no written interpretation**, and **absolute file paths** that only run on your own machine. Restart the kernel and run all before submitting.

Narrative

Lead with the finding, not the method: "Glucose separates the classes" beats "we plotted glucose". One figure, one message — if a plot needs two sentences to explain, it is two plots. Sequence the report: what the data is → what is wrong with it → what it shows → what you cannot conclude. Annotate directly on the figure. State limitations in the same breath as the finding.

A **dashboard** answers recurring questions; a **report** answers one question well. Know which you are building before you start.

KEY TERMS

`Grammar of graphics` `Aesthetic mapping` `Geometry` `Facet` `Small multiples` `Truncated axis` `Perceptually uniform colour map` `Data-ink ratio`

COMMON ERRORS TO AVOID

- Truncating a bar chart's y-axis
- Using a pie chart to compare more than three categories
- Leaving a legend to do work that direct annotation would do better
- Publishing a figure with no sentence stating what it shows

READINGS — ALL FREE TO ACCESS

1. Wilke — Fundamentals of Data Visualization
   `https://clauswilke.com/dataviz/`
2. CMU 36-315 — Statistical Graphics and Visualization
   `https://cmu-36315.netlify.app/`
3. seaborn tutorial
   `https://seaborn.pydata.org/tutorial.html`

TEST YOURSELF

1. Name the six layers of the grammar of graphics.
2. When is a truncated y-axis defensible?
3. You have two categorical variables with six levels each. Which plot, and why?
4. Rewrite 'Figure 3: PageValues by Revenue' as a title that states the finding.